

tronics test and measurement instrument manufacturers have been trying to extend the range of measurement in a way that it can be used for more applications. But, with increased range of measurement, there comes an increased complexity of design. This should not compromise other features. Every component has to become smaller and more components have to be integrated without increasing the probability of leakages. Rohde and Schwarz (R&S) FSH handheld spectrum analyser series now offers a frequency range up to 20GHz, and Keysight FieldFox series RF and microwave analysers have models that cover a frequency range of about 26.5GHz, thereby covering most aerospace and defence applications.

Smaller size. Technologies like quad flat no-leads (QFN) packaging have helped reduce the size of individual components and, hence, the device as a whole. In this technology, integrated circuits (ICs) are mounted on PCB surfaces without leads or through-holes. Another method of size-reduction used in the radio frequency range is by using monolithic microwave ICs (MMICs) instead of discrete components for functions such as power amplification, microwave mixing, and low-noise and high-frequency switching. This not only reduces the size but also lowers the power consumption, thereby improving battery life. The new RSA306 spectrum analyser by Tektronix is as small as 3.05x19.05x12.7cm and weighs just about 0.59kg.

Integration of devices. Integrating multiple instruments into a single system is a major trend we now see across different verticals of the industry. Handhelds usually have several devices integrated into a single system with small form factor, and the cost is also much lower compared to benchtops. A single handheld instrument occupies very little space compared to benchtops. MW82119B PIM Master by Anritsu combines PIM

Features of Different Handheld Devices

Product	Model	Vendor	Features
Base station analyser	BTS Master MT8220T	Anritsu	Advanced support for 4G wireless networks as well as installed 2G, 3G and WIMAX networks. Combines two-port cable and antenna analyser, spectrum analyser, power meter (with external power sensor), GPS receiver with antenna, bias tee, interference analyser with mapping channel scanner, vector signal generator and signal analysers.
Cable and antenna analyser	Site Master series	Anritsu	Provides the widest frequency coverage of 1MHz to 40GHz. Features vector network analyser (VNA) measurement functionality. Designed for measuring coaxial and waveguide systems. Conducts measurements such as return loss, VSWR, cable loss, DTF, phase, Smith chart, two-port transmission measurements and two-port cable loss tests.
Modulation analyser	LMR Master S412E	Anritsu	Targeted for land mobile radio (LMR), professional mobile radio (PMR) technicians and engineers in the field. TETRA analyser and TETRA coverage mapping options included. TETRA summary display to view data such as base and mobile colour codes, mobile network code, location area code, and mobile station maximum transmit power level.
Passive intermodulation (PIM) analyser	PIM Master MW82119B	Anritsu	Integrates 40W battery-operated PIM analyser, 2MHz to 3GHz cable and antenna analyser. Supports PIM versus time, swept PIM, distance-to-PIM (DTP), return loss, VSWR, cable loss and distance-to-fault (DTF) measurements. Rugged, with big display and intuitive UI.
RF and microwave analyser	FieldFox series	Keysight	Maximum frequency range of 4GHz up to 26.5GHz. Application-specific modes of operation, with software upgrade option. Features like QuickCal, InstaAlign and CalReady.
Spectrum analyser	FSH13 and FSH20	Rohde & Schwarz	Equipped with an amplifier that can be activated. Extended frequency to microwave range (9kHz to 13.6GHz and 9kHz to 20GHz). Features like LTE –advanced carrier aggregation.
Spectrum analyser	RSA306	Tektronix	Works in real-time, broad frequency range (9kHz to 6.2GHz), 40MHz real-time bandwidth, compact and weighs about 0.59kg. Comes with free SignalVu-PC software that provides advanced RF signal analysis capabilities. Less than half the price of conventional spectrum analysers.
Transport tester	Network Master Pro MT1000A	Anritsu	All-in-one network field tester. Supports high-speed fibre channel systems and expanded Ethernet diagnostic capabilities. Aimed at technicians who install and maintain mobile-access, fixed-access, metro and core transmission telecoms networks.
Transport tester	Network Master Flex MT1100A	Anritsu	Supports testing from 1.5 Mbps to 100Gbps. Four 100Gbps ports, each of which can be used independently and simultaneously for network and network-equipment tests. Supports OTN, Ethernet, SDH/SONET, PDH/DSn, and fibre channel in field, lab, manufacturing, and research and development environments.